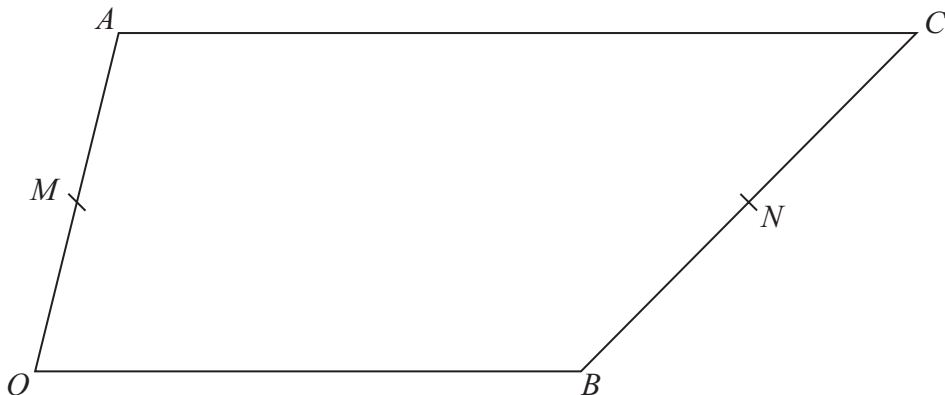


19 The diagram shows quadrilateral $OACB$.



M is the midpoint of OA .

N is the point on BC such that $BN:NC = 4:5$

$$\overrightarrow{OA} = \mathbf{a} \quad \overrightarrow{OB} = \mathbf{b} \quad \overrightarrow{AC} = k\mathbf{b} \text{ where } k \text{ is a positive integer.}$$

- (a) Express $\overrightarrow{MN}$ in terms of k , $\mathbf{a}$ and $\mathbf{b}$.

Give your answer in its simplest form.

(4)

- (b) Is MN parallel to OB ?

Give a reason for your answer.

(1)

(Total for Question 19 is 5 marks)